FINRAG - Agentic Retrieval Augmented Generation based Financial Assistant using Llama Index and Langchain Agents

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***Abstract*— In the financial world, Artificial Intelligence (AI) based tools such as the Expense Dashboard put forward great solutions for financial assistance. By using AI in financial technology, these platforms give users a better picture of their spending, which in turn helps them make better informed financial decisions. The Expense Dashboard in this work uses the platform of FINtech Retrieval Augmented Generation (FINRAG) which puts together what Large Language Models (LLMs) do best with the ability to retrieve data as needed. This is made available to our users in the form of relevant and flexible financial advice which plays out very well in in depth financial analysis. Also, by working with large sets of financial data, this work puts together personal tips for our users on budgeting, tax savings, and also compliance with current and changing financial regulations. The present AI enabled dashboard also steps out of the basic data report and into the world of Natural Language Processing (NLP) which is used to analyze what the user is looking for and present them with that info in a visual format. For instance, out of the box this work includs features which allow users to choose from which financial data sets they would like to see reports from which they upload as CSVs and within seconds have at hand bar, pie, and line graphs. Also, these visuals in real time help people and companies to see what they are spending and in turn to make better financial decisions. With this secure data management features, this AI platform protects user's sensitive financial info at the same time which also able to give timely and personal guidance. This research work also has machine learning algorithms which present predictive info to user thus helping you to better plan your finance and tax reports. From tax savings for individuals to record keeping for businesses to real time financial advice are offered by this AI powered systems transform. As the technology grows, these platforms become integral tools which makes financial management not only easy but smart and efficient.**

Keywords*— *artificial intelligence, machine learning, large language models, natural language processing, retrieval augmented genertion

I. INTRODUCTION

Managing an individual's or a company's finances involves providing fast answers, recommendations tailored to their needs, and proper handling of sensitive information. At the crossroad of AI and Fintech, Retrieval Augmented Generation (RAG) stands out as a powerful approach, as it empowers intelligent financial assistants to provide users with tailored advice. Integrating these kind of models with real-time data retrieval enables contextually relevant and up-to-the-minute financial guidance. RAG's ability to provide comprehensive answers combines with incessant user queries, navigating large sprints of financial data.

In financial applications, we see that RAG based systems use the best of what Natural Language Processing (NLP) and machine learning have to offer to analyze large sets of financial data, automate the decision making processes, and put forth predictive financial insights. For example in tax management which is a key area we see that these systems are able to not only go over an individual's financial data but also to find out the latest in tax regulations or tax saving options which they present in real time.

These systems' approach is to get to the point info from external sources (which may be financial reports, tax laws, real time market data) and at the same time use the in depth knowledge of LLMs to put together practical insights. Whether it is in the area of tax return preparation, finding out which budget cuts will work best, or making sure that our clients are in compliance with financial rules RAG does it all very accurately, dynamically and to the individual's needs.

Further, RAG structured virtual financial aides may enhance productivity in areas that are slow and riddled with problems. To give an example, accountants and businesses are able to automate menial tasks, like drafting taxes and tagging expenses which saves time and is remarkably accurate. Empowered users are able to make decisions with the information presented to them through guided explanations, recommendations, and insights while having the ultimate control. By continually incorporating new knowledge and learning from data, AI-powered financial assistants based on RAG are set to become indispensable tools in both business and personal finance. They are revolutionizing the way financial planning, tax compliance,

and compliance are managed, making them more efficient, accurate, and accessible to all users

II. Related Works

This section explains elaborately explains about the existing works, their insights, advantages and disadvantages in the relevant area of study.

AI for Predictive Analytics in Financial Management [1] discussed discusses how AI improves the accuracy of financial forecasting, enhancing tax planning and compliance strategies. Predictive analytics enables improved decision-making through pattern recognition in financial data. AI-based predictive analytics, unlike conventional models, adjusts dynamically to changes in financial data, enhancing tax compliance decision-making by minimizing human error and maximizing financial results.

Comprehensive AI-Enabled Financial Domain Review [2] presented a broad review of AI applications in finance, covering machine learning, NLP, and automation. The paper provided an extensive analysis of AI's impact on fraud detection, financial forecasting, and decision-making. While comprehensive, this study lacks emphasis on tax-specific applications compared to our AI-powered financial assistant, which is designed to provide tailored tax solutions and compliance recommendations using Retrieval-Augmented Generation (RAG).

AI and Financial Services [3] mentioned the preliminary research on AI in the financial sector, especially focusing on the advantages of automation. Furthermore, the study details how AI can optimize banking processes, customer relations, and increase the safety of finances. However, this was in its foundational stages when comparing existing frameworks to modern applications of systems like ours which utilize LLMs and Retrieval-Augmented Generation (RAG) for real-time tax compliance, thus addressing contemporary issues in burdened fiscal management.

Parallel Computing in AI-Based Financial Analysis [4] addressed the application of parallel computing in AI-driven financial decision-making. The research points out how AI models utilize parallel processing to process enormous amounts of financial data effectively. Although parallel computing improves speed and scalability, our method incorporates RAG for accurate document retrieval, further enhancing decision-making by ensuring the retrieval of the most appropriate tax law information in real-time.

Role of AI in Financial Management examined the use of AI in fraud detection and the effects on decision-making in finance management. Prof. Davis's research examines the use of AI on model implementations and transaction anomaly detection and activity prediction models [9]. While fraud detection is an essential piece of the puzzle, our work delves deeper into issues of compliance and deduction automation as well as optimization with the aim of allowing a taxpayer to legally save as much money as possible.

III. METHODOLOGY

This section explains how the proposed method could be used in the area of financial assistance to the users technically. This process starts when the user invokes a query, which is the basic input to the system. This query may vary from detailed financial queries to more general tax-related commands with context as a necessity. For example, for a system that deals with a tax filing system for finances, a user would query, "What are the tax deductions freelancers can claim?" In order to effectively address this, the system must correctly ascertain the user's intent using NLP to determine the most related tax information to return. Right interpretation of the question is needed in order to return correct as well as contextual answers.

When it gets the user query, the system embeds it through an embedding model, which converts the text input into a high- dimensional numerical representation called an embedding. The embeddings capture the semantic content of the query so that the system can see the context and intent and not just keyword matching. This allows for efficient and precise comparisons between the user query and the knowledge base stored. Here in this project, we use models like Google's Gemini API, BAAI's BGE (Bayesian Graph Embedding) model, and sentence- transformers from the Hugging Face library to create embeddings. Also, LlamaParse is used for document parsing so that it identifies useful insights out of documents with different formats such as PDF and Markdown. Such models utilize cutting-edge, pre-trained deep learning structures such as BERT or Transformer-based encoders, which have been fine-tuned on enormous, diverse corpora to manage sophisticated natural language understanding tasks.

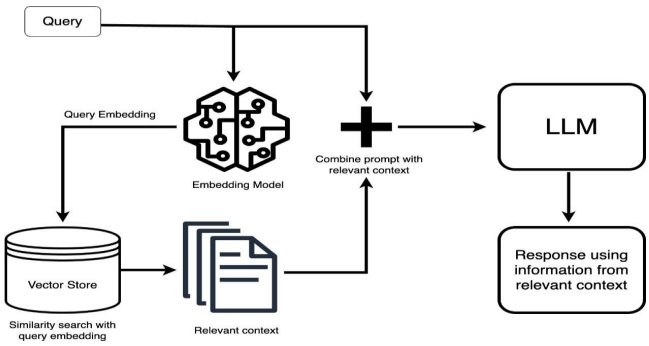


Fig.1. Overview of RAG

After the query embedding is generated, a similarity search in a vector store is performed using it. The vector store is a repository of precomputed embeddings of documents, books of records, policies, or structured data that the system looks up on a search. Unlike typical keyword search, the vector store makes use of advanced mathematical distances, such as cosine similarity or Euclidean distance, to assess the similarity of the query embedding with those contained in the store [30]. This helps the system be able to yield semantically equal documents or entries even when there is no direct match of the query with that stored text. In this project, LlamaIndex (previously GPT Index) is used as the vector store, which supports semantic search functionality. LlamaIndex caches

precomputed embeddings and offers a good method of performing similarity searches among the data with its vector store index and embed model integration like HuggingFace and Gemini. This configuration will guarantee retrieval of the most highly relevant results regardless of whether there are noisy or complex datasets involved, making it especially beneficial to query financial or legal data within tax filing systems and it is shown in Fig.1.

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Based on the similarity search, the vector store generates a list of most relevant documents or data snippets for the user query. These retrieved documents are the additional context to be utilized to generate a more informed answer. For instance, when the query is a tax deduction, the system can retrieve authoritative tax law documents, case studies, or guidelines related to self-employed taxpayers. The quality of the retrieved context is crucial in ensuring that the final response is credible and true [25]. The acquired contextual information relevant to the query is then combined with the original query to create a well-crafted prompt. This facilitates that when the query is passed to the Large Language Model (LLM), it gets supplemented with background knowledge. By providing the LLM with more context, the chances of generating a correct and complete factually accurate response increase. This prompt engineering technique is widely used by AI-driven applications to reduce hallucinations and improve answer accuracy, and it is shown Fig.2.

Lastly, the enhanced prompt is input to the LLM, which produces an output depending on both the query and the obtained context. As the LLM can now see the obtained documents, it can produce a more accurate and relevant response instead of using only its pre-trained knowledge. The produced response is then passed back to the user so that it will be aligned with the obtained documents [21]. Under a taxation return framework, such a method provides conformity to law and answers depend on the most current financial directives.

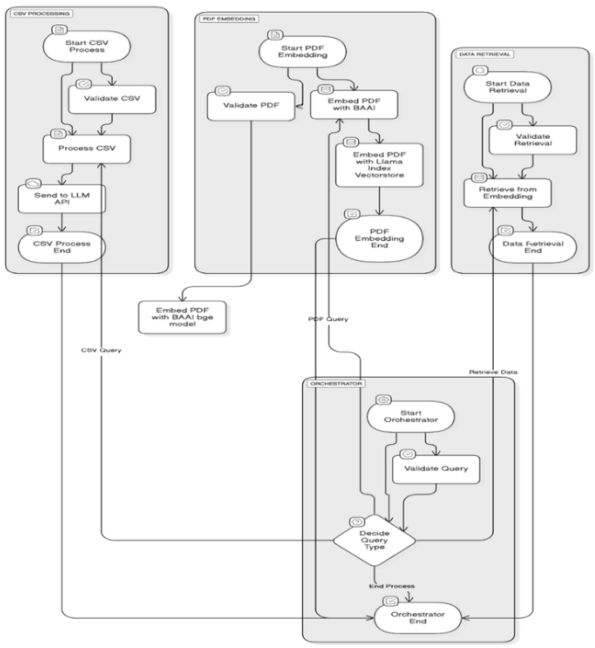


Fig.2. RAG Flow diagram

IV Results and Discussion

Both a generic LLM (GPT-4) and our agentic RAG-based LLM were given 50 queries on taxation and financial assistance. The responses generated by them were evaluated and compared using evaluation metrics.

TABLE 1: Comparison of generic LLMs with a RAG based LLM

Criteria	LLM only	LLM+RAG
Accuracy	May misinterpret laws	References exact context
Contextual awareness	Limited to prompt data	Uses retrieved context for grounding
consistency	May wary on phrasing	More stable and backed up by static embedded source
maintenance	No updates unless fine tuned	Easy to update just by changing context

RAG-based LLMs surpass regular LLMs in fiscal and legal operations by offering improved accuracy and context sensitivity through document retrieval. Regular LLMs can misread laws and lack consistency with wording, whereas RAG systems base responses on source material, thus offering more consistent and accurate results. RAG models are also simpler to upkeep, as it is easier to update context compared to fine-tuning. This makes RAG suitable for ever-changing domains such as taxation and finance.

Evaluation metreics which include precision, F1-score, recall which are briefly discussed and compared as follows.

It evaluates the ranking of the first relevant document in the retrieved results. It is defined as:

$$\mathbf{MRR = (1 / |Q|) * \Sigma\ (1 / rank_i),\ for\ i = 1\ to\ |Q|}$$

A higher MRR indicates that relevant documents are ranked higher in the search result, and the comparitve results showed in Table 2 and Fig.3.

TABLE 2: Comparison of performance between a baseline model and our RAG based model

Metric	LLM	RAG based LLM	Notes
Precision	0.60	0.95	LLM guesses some, while RAG returns mostly accurate responses.
Recall	0.55	0.93	LLM misses more valid answers; RAG catches more due to access.
F1 Score	0.57	0.94	RAG strikes balance between precision and recall.

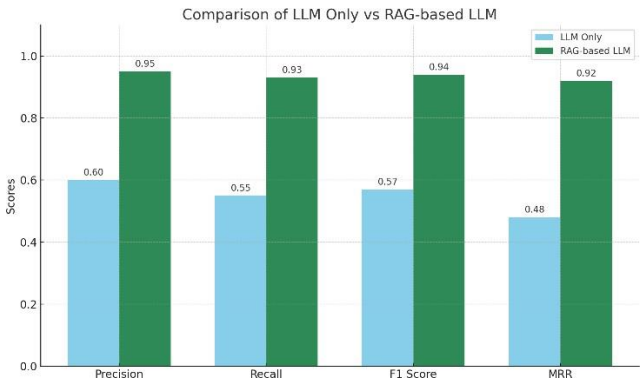


Fig.3. Comparison of LLM only vs RAG-based LLM

BERTScore is a semantic evaluation metric that uses pre-trained language models like BERT to assess the similarity between a predicted response and a reference answer. Unlike traditional metrics such as BLEU or ROUGE, which rely on exact word matches or n-gram overlap, BERTScore evaluates meaning similarity [13] by comparing contextualized token embeddings. The observation on calculating bertscore with 10 questions are tabulated below Table 3 and Fig.4.

TABLE 3: BERTScore of LLM vs FINRAG

Questions	RAG-Based LLM	Context-less LLM
1	0.94	0.74
2	0.92	0.71
3	0.97	0.83
4	0.91	0.68
5	0.95	0.77
6	0.90	0.66
7	0.96	0.78
8	0.93	0.70
9	0.94	0.76
10	0.91	0.69

ROUGE-L: Recall-Oriented Understudy for Gisting Evaluation – Longest Common Subsequence is a widely used evaluation metric for natural language generation tasks, particularly in summarization and question answering. Unlike metrics that rely solely on exact n-gram matches [15], ROUGE-L focuses on measuring the Longest Common Subsequence (LCS) between the reference and generated texts.The observation on calculating rouge with 10 questions are given below in Table 4.

TABLE 4: ROUGE-L of LLM vs FINRAG

Q	RAG-Based LLM	Context-less LLM
1	0.88	0.68
2	0.89	0.67
3	0.93	0.79
4	0.86	0.61
5	0.90	0.73
6	0.85	0.61
7	0.91	0.75
8	0.88	0.65
9	0.89	0.72
10	0.87	0.63

To find relevant financial documents, the BAAI BGE-based retrieval system uses cosine similarity between the query embedding.

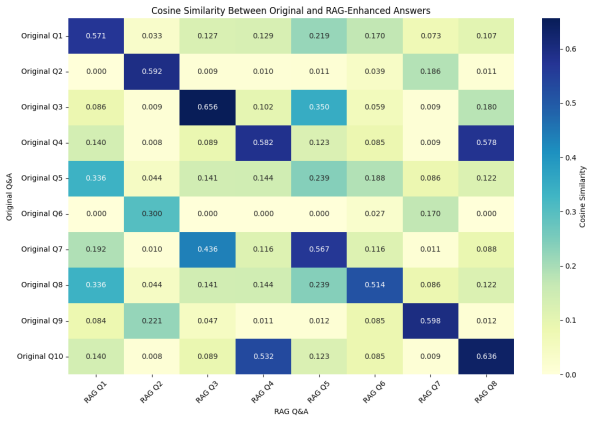


Fig.4.Cosine similarity on CSV querying

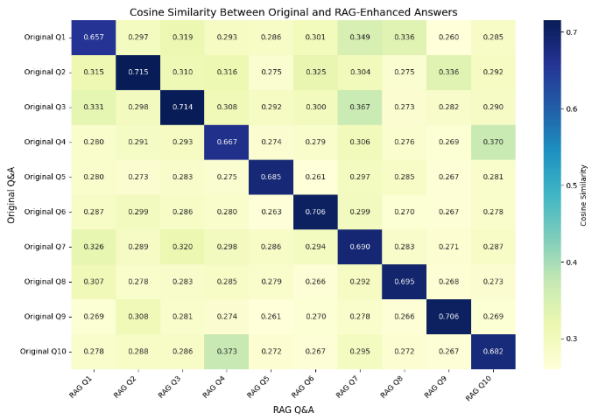


Fig.5. Cosine similarity on Tax querying

V.CONCLUSION

This project constructs end-to-end AI-powered financial assistant to fill the gap between personal cost management and complex tax law interpretation. From CSV data input with PostgreSQL integration, natural language processing using Gemini, and tax question answering using a Retrieval-Augmented Generation (RAG) approach.

The application of LangChain agents makes dynamic tool-switching possible on the basis of the type of user query, showcasing the prowess of modular reasoning-based systems. The Streamlit-based interface provides a smooth and interactive user interface, bringing powerful technologies within the reach of end-users without having to possess technical expertise.

This solution improves not only the efficacy of tracking and reporting expenses but also engages users to make better decisions by querying taxation regulations in a natural way. It provides the basis for AI-based financial applications in the future, including potential extensions into investment guidance, income prediction, and regulatory compliance.

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